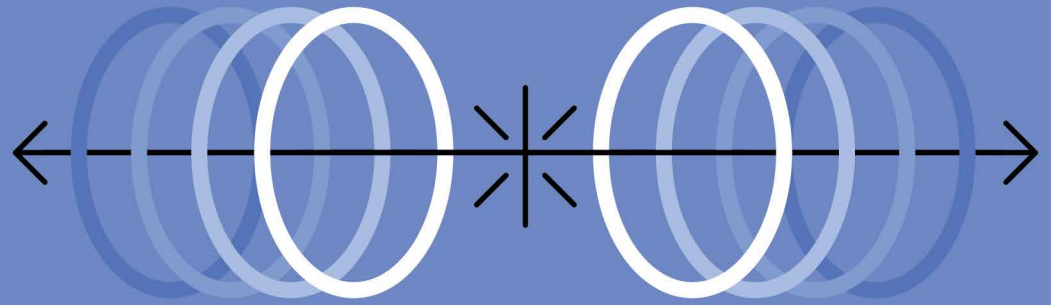




Redefining the Back Office with AI

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Orchestrating the Back Office into a Competitive Frontier

For decades, back-office operations were viewed as a necessary cost—essential to running a business but rarely a source of strategic advantage. This perception is now outdated. Artificial intelligence (AI) is changing that narrative, fundamentally redefining what back-office functions can do, how quickly they can do it, and what it means for an organization's ability to compete. The research in this paper, sponsored by TP in association with Harvard Business Review Analytic Services, arrives at a pivotal moment: The window for meaningful transformation is open, but it is not open indefinitely.

The urgency is real. Organizations already expect AI to significantly transform their workflows, and this change isn't gradual—it's a mandate for speed and scale that demands a fundamentally different approach to how companies think about their operations, their people, and their partners. Yet the greatest barriers are not technical. Organizational and cultural challenges such as data silos, change resistance, governance gaps, and skills shortfalls account for most implementation difficulties. Organizations that treat this moment as an opportunity to experiment cautiously need to balance the risk of finding themselves structurally disadvantaged against competitors that are building AI-native capabilities

while being thoughtful about implementations at scale. What makes this transformation genuinely different from prior waves of automation is not the technology itself but the trust it requires to work. AI in the back office does not simply replace manual steps. It changes who makes decisions, how those decisions are made, and how accountable the organization is for their outcomes. Employees must trust that AI will enhance their roles rather than eliminate them. Leadership must trust that AI-driven processes will meet the governance and compliance standards their industries demand. Building that trust is not a communications challenge; it's a design and implementation discipline that must be embedded from the outset, with human-centered principles guiding every stage.

This juncture is precisely where deep operational expertise combined with AI capability proves decisive. It's why TP developed TP.ai FAB (foundational AI backbone), a modular platform that orchestrates people, technology, and industry-specific expertise into integrated back-office operations. By combining vertical-specific solutions with intelligent AI orchestration and a secure foundation, TP.ai FAB is designed to accelerate time-to-value while preserving the governance and human oversight that regulated industries require and building the



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extensibility needed for the next wave of autonomous systems.

At TP, we believe the back office is more than a support function waiting to be modernized—it's a strategic asset waiting to be unlocked. The AI-driven back office will reduce friction in the customer experience, augment and elevate the service representative's experience, and deliver with speed and efficiency. The organizations that will thrive in the coming years are those that move with speed and intention and embrace transformation as a core competency rather than a one-time project. This paper provides valuable insights into that journey and we are proud to have sponsored it.

Redefining the Back Office with AI

Back-office functions in areas such as finance and accounting, account management, collections, sales, and HR have been modernized through artificial intelligence (AI) and emotional intelligence (EI). AI drives efficiencies across all back-office functions, while EI enhances both customer-facing operations and internal collaboration.

WHAT MAKES AI special is that it can “revolutionize” various back-office functions, “trimming costs, bolstering decision-making procedures, mitigating risks, and amplifying customer service,” according to a 2025 report, “Uncover Hidden Value Through Back-Office AI,” from London-based Deloitte Touche Tohmatsu Ltd. Generative AI (gen AI), in particular, excels at the document-heavy knowledge work characteristic of back-office operations, such as processing contracts, managing correspondence, and analyzing unstructured data across finance, HR, and procurement. The urgency of the transformation that AI can effect is reflected in executive expectations across industries. A 2025 survey by New York-based McKinsey & Co. found nearly half of organizations (47%) expect AI to change 30% or more of their workflows within just one year, while an additional 34% anticipate significant changes within one to five years. **FIGURE 1**

This compressed timeline reflects both the maturity of available AI technologies and the competitive pressure companies face to modernize their back-office operations quickly. The benefits of back-office modernization can be significant, including cost reduction, increased productivity, enhanced compliance, improved accuracy, better data management, and higher collection rates and sales conversion rates. But corporate executives are already looking for more. “Companies have shifted their focus from efficiency and cost savings to strategy and innovation,” says Anil Vijayan, a partner at the Everest Group, a global research firm headquartered in Dallas. “Operated correctly, the back office can create a huge competitive advantage.”

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HIGHLIGHTS

Artificial intelligence (AI) creates strategic impact by **converting back-office data into competitive intelligence that drives business decisions**, freeing skilled employees from routine tasks to focus on innovation and customer relationships.

The biggest impact of back-office AI may not be efficiency but **enabling employees to use their emotional intelligence**.

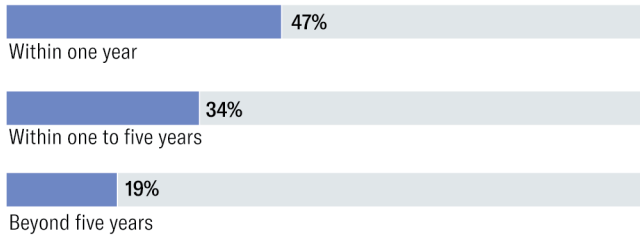
Organizations must build transformation capability as a core competency, treating each AI implementation not as a destination but **as preparation for the next wave of autonomous systems**.

FIGURE 1

An Artificial Intelligence Impact Timeline

Companies are anticipating fairly imminent workflow transformation from AI

Percentage of organizations expecting AI to change 30%+ of workflows



Source: McKinsey & Co. survey, January 2025

enabling faster responses to market changes than competitors can match.

However, achieving these strategic benefits hinges on a thoughtful implementation approach. Companies must address integration challenges and change management requirements and then decide whether to replace legacy systems or bolster them with modular enhancements. “Most organizations are not thinking enough about the big picture when they are undertaking AI-led back-office modernization projects,” says Simon Baxter, research director at TechMarketView, a British technology analysis firm covering the software and IT services market.

The technical implementation challenges, while significant, pale in comparison to the people and culture issues that must be addressed. “AI itself has become relatively user-friendly and plug-and-play,” Baxter says. “A lot of the challenges really come from people, such as resistance to change, business culture, complex existing processes, and governance issues.” This insight reveals that competitive advantage will increasingly flow to organizations that excel at human-centered change management rather than technical sophistication alone.

A key part of meeting these challenges is selecting and working with operational service experts, who can help optimize processes and/or operate back-office functions. “Most organizations don’t realize they’re already benefiting from AI transformation through their existing vendor relationships,” says Nihar Malali, a technology executive at National Life Insurance in Dallas and author of *Digital Transformation in the Age of AI*. “We don’t have to implement AI internally, but all our vendors have integrated AI into their platforms—from office management tools to quality assurance systems. Using

existing vendor AI features significantly reduced our development costs and improved our operational capabilities.”

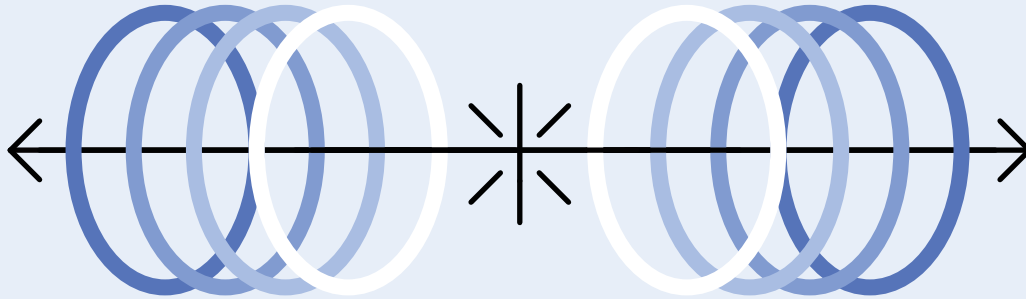
This report explores how companies can digitally enhance their back-office operations by seeking guidance from and partnering with operational service experts while maintaining human-centered design principles to enhance efficiencies, reduce costs, drive innovation, and achieve business goals. The report provides guidance for measuring success beyond traditional ROI, selecting strategic service partners, and building transformation capabilities for continuous technological evolution.

Implementation and Technical Realities

Back-office operations typically work in separate departments—finance, HR, operations, and customer service—with each using different systems and processes. This fragmentation prevents organizations from seeing whether they’re delivering services accurately, on time, or cost-effectively. Real-time integration across functions enables companies to make data-driven decisions and fix problems before they affect customers.

Kate Leggett, an analyst at Forrester Research in Cambridge, Mass., says the back-office transformation should provide seamless information flow between operational activities and decision making. “When a salesman is talking to a customer, they need to know if there’s a warranty claim or a part on back order,” she says. “This real-time integration prevents customer defection and identifies upselling opportunities.”

The fundamental technical challenge with implementing digital transformation involves achieving seamless data flow and process integration without completely replacing existing systems. Companies often face deeper structural obstacles that complicate AI adoption. As Leggett notes, “Enterprise-wide views are expensive and difficult to create because business functions hoard and accumulate highly customized data.”



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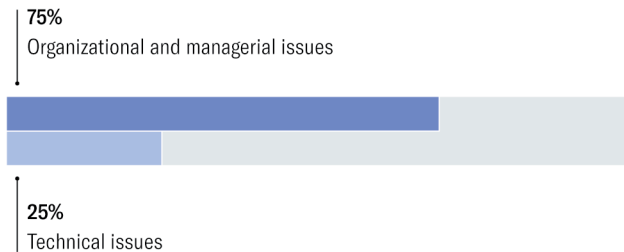
Nihar Malali Technology executive National Life Insurance	
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FIGURE 2

Digital Transformation Challenges

Organizational and managerial difficulties outweigh technical issues

What are the greater challenges to automation and digital technologies?



Source: London School of Economics survey, 2024

A 2024 survey by the London School of Economics,¹ based on over 1,200 automation and digital transformation cases, revealed that technical issues represent only 25% of implementation challenges, while organizational and managerial factors account for 75%. Companies face what the study terms “multiple organizational silos” that inhibit technology adoption and create culture, process, legacy system, data management, strategy, and skills gaps.² **FIGURE 2**

“These decisions involve multiple considerations beyond technical requirements,” Everest Group’s Vijayan explains. “First, what is the sensitivity or criticality of that process? Is it core to what you do and therefore necessary to keep in-house? Second, there’s a cost consideration which always comes into play.” This framework helps firms evaluate not just what can be automated but also whether the automation should be done in-house or outsourced.

While full replacement of legacy systems is often cost-prohibitive, organizations can take incremental approaches that preserve existing investments.

“Legacy systems don’t have to be barriers to back-office modernization,” National Life’s Malali explains. “Even at the largest banks, if you look at all the sophisticated mobile interfaces on the front end, the back-end systems are still running on mainframe technology.” Organizations can modernize by adding new layers that create updated interfaces while maintaining stable core systems.

However, these architectural approaches require careful prioritization decisions. As Malali explains, “The integration is the key thing. Are you looking for speed? Are you looking for performance, accuracy, and consistency? You can only do so much of the architecture and design to make all three of these [latter] elements available at any point in time.”

Looking ahead, the advent of agentic AI—autonomous systems that can execute multi-step workflows and make

decisions within specific parameters—will redefine a wide range of business processes. “Agentic AI has the potential to automate a large number of day-to-day tasks, which will only accelerate as the technology matures,” TechMarketView’s Baxter says. “There are two components for it to really take hold. The first is a focus on the right use cases by organizations, specifically ones where agentic AI can deliver immediate business value. The second is enterprise data readiness, which can be a large barrier to success. If your data is not accessible, of the right quality, or fit for purpose, then any AI project is not going to deliver the desired productivity gains.”

The Human-Centered Design Imperative

The most critical factor distinguishing successful back-office AI deployments involves maintaining a focus on human needs throughout the transformation process. Companies should maintain laser focus on when automation enhances human abilities versus when it degrades customer experience and employee satisfaction.

“I’ve yet to bump into someone who said, ‘That was an excellent piece of automation. I’ve just had a great experience with a chatbot,’” says Karen Cham, a professor of augmented intelligence, digital transformation, and design at Kingston University London. This experience gap comes from automating processes without considering the full customer journey or employee workflow. Organizations often implement isolated automation without understanding its integration or impact on humans, creating what Cham calls “artificial stupidity,” meaning systems that perform poorly in terms of human interaction.

Automation success varies by application type. “Customer-facing tools like chatbots struggle with context and emotional complexity, while internal process automation handles repetitive rule-based tasks well,” Cham says. Productivity gains mainly come from back-office operations—data

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processing, document routing, and compliance checks. Recognizing this balance helps companies focus on automation where it adds value and avoid areas that hurt the experience.

Effective automation keeps humans in control. “If you went into finance or HR and said, ‘We’re going to cocreate our automation,’ they would immediately know exactly what needed to be done,” Cham explains. Subject-matter experts understand nuances and exceptions that automated systems cannot handle.

Poor automation can cause ethical problems, such as bias against certain groups in credit decisions, harming customers and risking regulatory noncompliance. These issues often show up as opaque algorithmic decisions, weak integration between billing and customer service, and a lack of explainable AI for customer-facing systems, according to the experts interviewed for this report.

Regulated industries must balance automation with human oversight. Financial services, health care, and insurance need explainable decision making for compliance and customer protection. These requirements often favor hybrid models where AI provides data analysis and recommendations and humans retain decision authority, protecting companies from algorithmic bias while maintaining the processing speed and consistency that AI enables.

Understanding employee resistance is key to successful adoption. “I see people holding on to manual processes to protect their jobs,” Cham says. This fear of displacement, not opposition to better tools, drives resistance. Effective change management addresses these concerns and shows how automation can enhance job security.

Overcoming resistance requires showing employees personal value, not just organizational benefits. As Dave Ulrich, professor emeritus at the University of Michigan Ross School of Business in Ann Arbor, Mich., explains, “Commitment to change comes when employees see what’s in it for them, such as how AI activities will help employees reach their goals.” Focusing on employees ensures transformation aligns with career development and job satisfaction rather than being imposed from management above.

Enabling Human Emotional Intelligence

The biggest impact of back-office AI may not be efficiency but enabling employees to use their EI. A 2024 Massachusetts Institute of Technology study, “The EPOCH of AI-Human-Machine Complementarities at Work,” found that over the past decade companies have prioritized jobs that emphasize empathy, creativity, and leadership while removing routine tasks.

AI and EI can work together rather than compete against each other. For example, AI supports back-office claims collection by handling routine tasks and data insights, while humans add empathy and judgment, enabling payments to be recovered efficiently without harming customer relationships.

Yet, while companies recognize the potential of AI to amplify human skills, translating intention into action remains a challenge. According to Deloitte’s 2024 Global Human Capital Trends survey of 14,000 business and human resources executives across 95 countries, 71% of respondents report that their generative AI plans include advancing human capabilities such as empathy and curiosity. Yet only 9% say their companies are actually making progress toward that balance. **FIGURE 3**

The wide gap between intent and progress shows why AI must be designed to enhance human capabilities. This shift is changing workforce composition. In contact centers, Forrester’s Leggett says, “Agents will be more highly skilled, trained, and compensated. Lower-tier agents will manage automation, such as supervising chatbots or creating self-service content.” This evolution demonstrates that AI, when deployed thoughtfully, creates new categories of skilled work rather than simply eliminating positions. At the same time, AI can promote higher job satisfaction because the tools unburden employees from mundane manual tasks like preparing call/interaction summaries.

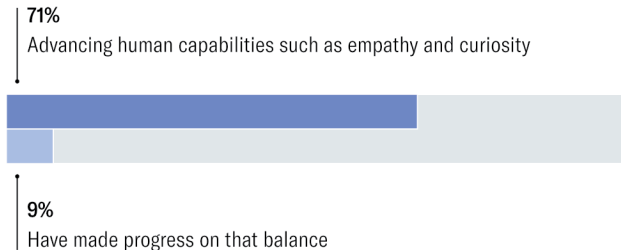
Executives recognize AI’s potential to enhance human skills, but turning that vision into action is difficult. Vijayan

FIGURE 3

Generative AI Plans Highlight Human Abilities

Companies seek more empathy and curiosity, but progress on the balance has been slow

Companies' plans for generative AI include:



Source: Deloitte survey, February 2024

notes that top companies now see back-office functions not just as efficiency tools but also as opportunities for “strategy and innovation.” AI removes routine burdens, letting teams strategically focus on customer needs, innovation, and relationship-building that drives long-term value.

Kingston’s Cham says this evolution needs intentional design, making employees “automation architects rather than passive recipients.”

Partnership Strategy

Most companies are not developing AI functions internally, experts report. Instead, they’re benefiting from AI integration through their existing vendor relationships, such as quality assurance systems with built-in conversation guidance. Forward-looking companies are developing strategic partnerships with business process services (BPS) providers that design, implement, and manage comprehensive multivendor solutions.

Baxter advises, “The best place to look is at who you’re already working with and see what they’ve got on the table.” Companies should evaluate vendors not only for AI capability but also for their commitment to innovation and change management.

Experts note the BPS industry has evolved from process optimization to technology-enabled transformation that systematically integrates AI adoption with workforce development and governance frameworks. As technologies like agentic AI emerge, this assessment must include not just current capabilities but also partners’ roadmaps for autonomous systems and their governance frameworks for managing increasingly complex AI deployments.

“Not all vendors are created equal,” Malali notes. “Organizations need partners that understand industry dynamics, regulations, and change management, not just ones that provide technology.” Strong partnerships go beyond software delivery to share risk, invest in outcomes, and collaborate on process optimization.

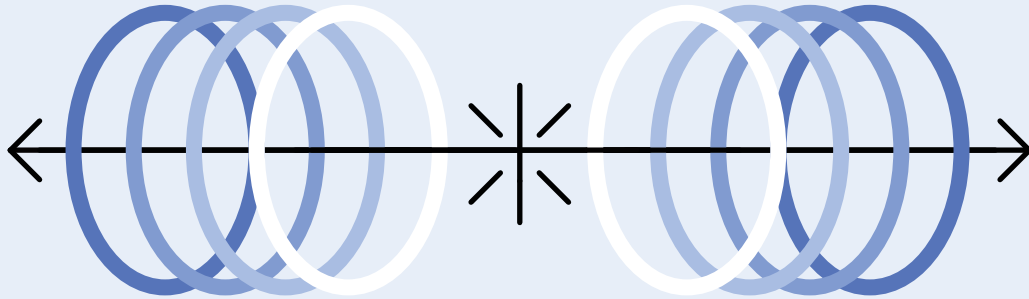
This approach to strategic partnerships becomes even more critical as AI development accelerates. Baxter’s research shows that U.K. organizations jumped from running five to 10 AI trials in 2023 to hundreds in 2024, requiring partners that can support rapid experimentation while maintaining governance standards.

Building Future-Ready Operations

The rapid evolution of AI technologies demands that organizations build transformation capability as a core competency rather than treating each implementation as a one-time project. Process-centric organizations, particularly those in the BPS sector, have developed this capability over years of technology adoption, building frameworks for continuous improvement that position them to integrate emerging technologies more systematically.

Such extensibility is necessary because organizations that optimize only for current AI applications risk being locked into approaches that can’t accommodate the next wave of innovation. “Design should always allow extensibility and change,” Malali says.

This architectural adaptability becomes particularly critical as the industry advances from traditional AI applications to agentic AI systems. The BPS industry has leveraged AI in back-office operations for years, using machine learning for data extraction, predictive analytics for workload management, and intelligent automation for process optimization. Agentic AI represents the next phase: systems that independently handle complex tasks end to end with minimal human intervention. In back-office operations, AI tools now handle routine tasks—such as invoice processing, insurance



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Dave Ulrich Professor emeritus University of Michigan Ross School of Business	
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“Measuring cultural readiness involves assessing employee engagement with new technologies, speed of adoption for new tools, and the capability to experiment and learn from failures.”

Kate Leggett, analyst, Forrester Research

claims triage, order updates, and shipment tracking—while humans intervene mainly for exceptions or judgment calls.

“Agentic AI has the potential to fundamentally change how we think about modernization,” Vijayan says. “It could eliminate the need for certain large enterprise systems we rely on today, which means the way we approach modernization may look completely different within three years.”

Baxter emphasizes that successful agentic AI adoption will require robust governance foundations. “Implementing clear usage policies, ethical practices, compliance and regulatory frameworks, and tools to monitor AI model token consumption will be crucial for organizations to manage the risks, capabilities, and cost of AI programs,” he says.

The most successful companies create what Leggett calls a “change-ready culture” that views continuous technological adaptation as a competitive advantage rather than a burden. “Measuring cultural readiness involves assessing employee engagement with new technologies, speed of adoption for new tools, and the capability to experiment and learn from failures,” she says.

This cultural foundation becomes essential as agentic AI systems begin executing workflows that previously required human coordination across both back-office and customer-facing operations. Organizations must build transformation capability as a core competency, treating each AI implementation not as a destination but as preparation for the next wave of autonomous systems.

Conclusion

The future of back-office transformation demands a fundamental reimagining of how organizations approach technological change and competitive positioning. As Baxter says, “You now have AI-native companies that are fully embracing this technology [gen AI] and building their businesses to try and disrupt incumbent suppliers.”

Organizations that successfully integrate AI into their back-office operations create compounding advantages that

become increasingly difficult for competitors to replicate. They develop change-ready cultures, strategic partnerships that drive innovation, and measurement frameworks that capture both immediate gains and long-term strategic value. Most important, they position human EI as a differentiating capability rather than treating technology as a replacement for human judgment.

Success requires treating transformation as an ongoing capability rather than a one-time project. As Cham underscores, positioning employees as cocreators of automation ensures their expertise guides implementation and validates outputs. Organizations must plan for perpetual change, building transformation capability as a core competency.

The back-office transformation represents more than operational improvement. It signals the emergence of sustainable competitive advantage where AI efficiency combines with human EI to create market differentiation that competitors cannot easily replicate. The organizations that will lead this transformation are those that combine internal vision with external expertise, using strategic solution partners that bring both technical capability and operational experience to accelerate success and ensure sustainable outcomes.

According to Baxter, “You’ve got to think more broadly about existing processes, people, and business outcomes and tie that all together with the goals for implementing an AI tool.”

Endnotes

- 1 Leslie P. Willcocks, John Hindle, Matt Stanton, and John Smith, *Maximizing Value with Automation and Digital Transformation: A Realist's Guide* (Cham, Switzerland: Palgrave Macmillan, 2024), <https://doi.org/10.1007/978-3-031-46569-7>.
- 2 Ibid.



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